Simulating Logistics Systems

Lesson 6/15

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Outline of this lecture

Data processing and analysis

Generation of random numbers

Generation of random variables

Adherence Tests

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Data processing and analysis

Generation of random numbers

Generation of random variables

Adherence Tests

Collection and use of sample data

Most systems have **random** data sources that need to be *appropriately* represented. The goal is to obtain probabilistic models that allow us to make *inferences* about the properties of a given random phenomenon. These are the **input models** used in the simulation.

The data can be used in the following ways after collection:

1. *Use the values recorded in the past*.
2. Use the data to define an *empirical* distribution.
3. Use the data to make statistical inferences aimed at finding the best theoretical probability distribution fit (**data modeling**).

There are problems related to approaches 1 and 2.

- Approach 1 repeats the past.
- Approach 2 only generates data within the observed range with irregular steps.

Collection and use of sample data

It is **preferable** to represent data using a theoretical probability distribution because:

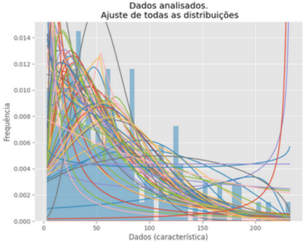
- Empirical distributions can have irregular "steps."
- Empirical distributions only generate data within the observed range.
- Some phenomena tend to follow predefined theoretical distributions.
- A compact representation of the theoretical distribution facilitates manipulation and modification (*e.g.: the exponential distribution E(m) only needs one parameter, "m," the mean, to be configured*).

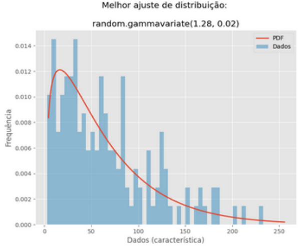
Using a theoretical distribution may not be possible in some situations.

- Data from a mixed group (e.g., data from new and old trucks);
- Values can't be too high or low (e.g., normal distribution → solution: truncate).

Data Modeling Steps

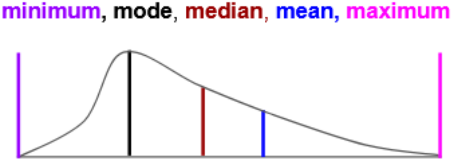
1. **Data Collection (Sampling)**: Fieldwork: Use 100–200 data points. Collect data in order of occurrence (for correlation analysis). Be mindful of seasonality.
2. **Data Processing (Describing the Data)**: Use statistical software to analyze unusual values due to collection errors or rare events.
3. **Inference: Inferring the behavior of the population**: Use statistical software and programming routines. Determine the best probability distribution for the data (probabilistic model).





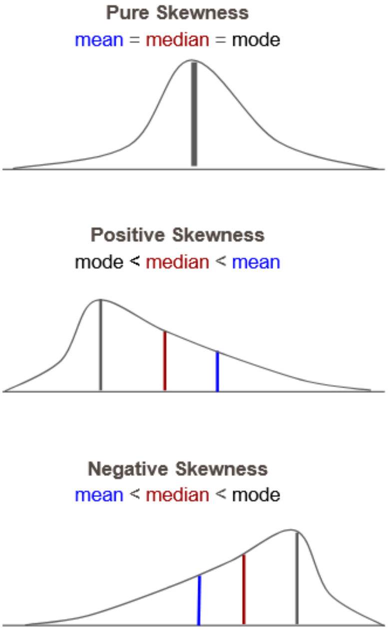
Sample measures (position measures)

- **Mean**: The arithmetic mean of the data values (equivalent to the center of mass). $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$
- **Median**: The central observation *after the data has been sorted*.
- **Mode**: The *most frequent* value in the sample.
- **Minimum** and **maximum**: the *extreme values* in the sample.



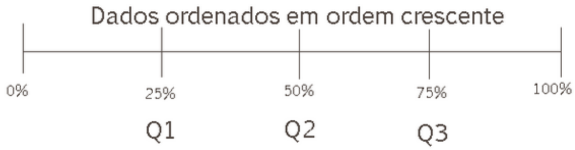
Sample Measures (Measures of Dispersion)

- **Range**: Highest less lowest value.
- **Variance**: $s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$
  (*how much the data deviates from the mean*).
- **Standard Deviation**: $s = \sqrt{s^2}$
  (the same unit as the sample).
- **Coefficient of variation**: $v = \frac{s}{\bar{x}}$
  (*measures relative variability*).
- **Skewness**:
  - Pure: mean $=$ median $=$ mode
  - Positive: mode $<$ median $<$ mean
  - Negative: mean $<$ median $<$ mode



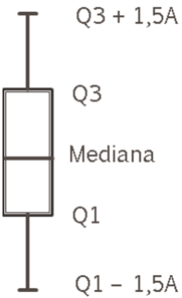
Sample Measures (Measures of Dispersion)

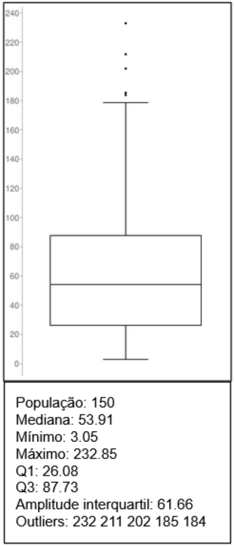
- **Quartiles**: divide the data into four equal parts.
- **Order p quartile** $(0 \leq p \leq 1)$: The value that divides the sample into two parts such that the proportion p of the sample data is to the left of X_p and the proportion $(1-p)$ is to the right.
- **Order p percentile** $(0 \leq p \leq 100)$: Same as quartile, but expressed as a percentage.



Sample Measures (Measures of Dispersion)

- Outliers: Values discrepant from the sample.
  - Moderate: $Q_1 – 1.5A \leq v \leq Q_3 + 1.5A$
  - Extreme: $Q_1 – 3A \leq v \leq Q_3 + 3A$
  - A: $= Q_3 - Q_1$: Range inter-quartile
- Box Plot:





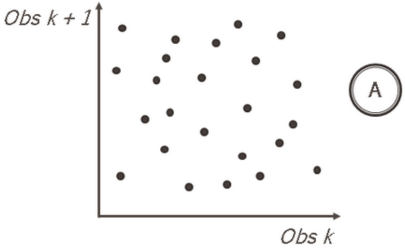
Dependency/independence between data

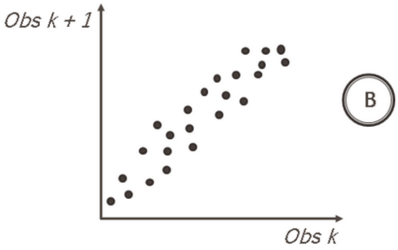
Check for dependencies between variables. *Are there correlations between the observations that make up the sample?*

- Examples of independent variables:
  1. Variation in service time in relation to customer height.
  2. At a car wash, cleaning time varies depending on the time of day.
- Examples of dependent variables
  1. Service time increases with the passage of working time.
  2. Cleaning time varies with the size of the car at a car wash.

Correlation between data

Check whether the sample represents a sequence of independent and identically distributed (i.i.d.) values. Each point is formed by the pairs $(x_1, x_2), (x_2, x_3), (x_3, x_4) \dots, (x_k, x_{k+1})$

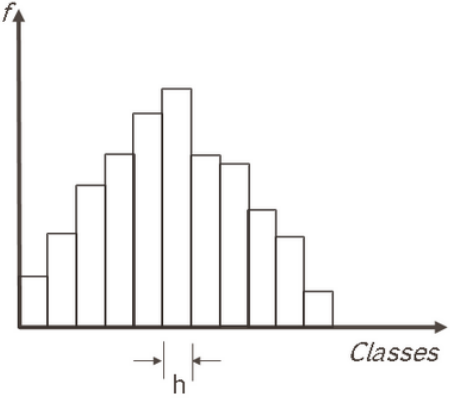




- A There is no correlation
- B There is correlation

Histograms

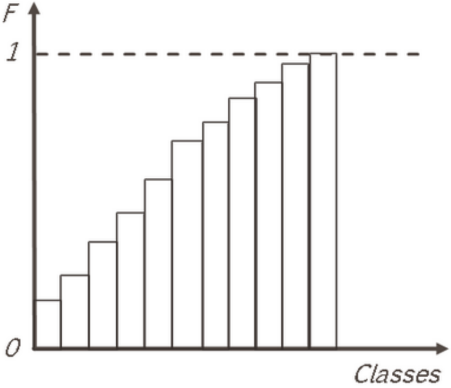
1. Shows how the data is distributed
2. f = frequency of each class $(0 \leq f \leq 1)$
3. $\%$ of samples in each class



- Number of classes (Sturges' rule): $k = 1 + 3.3\ log(n)$
- n = number of observed values
- Class width: $h = \frac{A}{k}$
- A = sample range

Histograms

- Cumulative distribution
- F = cumulative frequency of classes $(0 \leq F \leq 1)$
- Cumulative percentage of samples in each class



- Ex: For $n = 199$ observed values and recorded amplitude of $A = 43$.
- K = 1 + 3.3 $log_{10}199$ = 1 + 3.3(2.3) = 8.59 ≈ 9
- K = 9 classes.
- The width of each class h would be:
- $h = A/K$ = 43/9 = 4.8
- h = 4.8
- Next, record the relative frequency of the data for each class.

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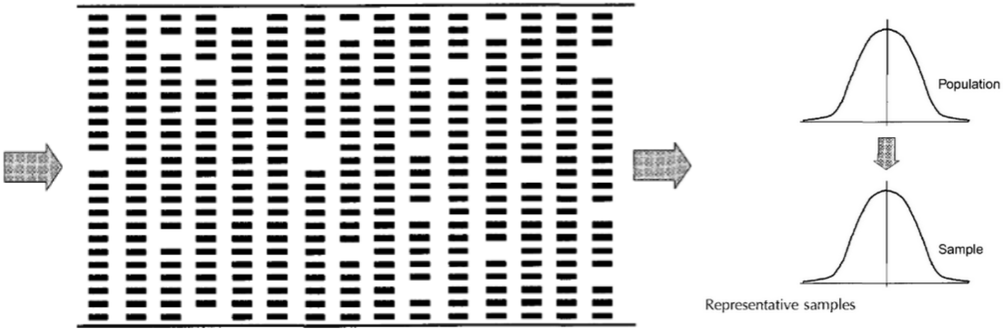
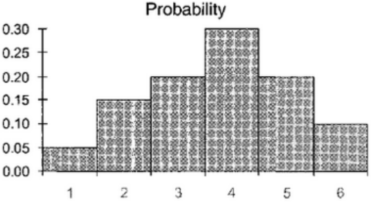
Generation of random numbers

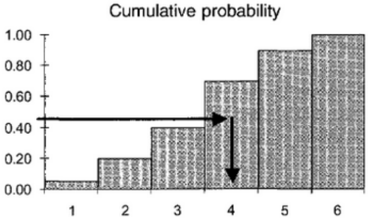
A conveyor belt carrying chocolate bars should be arranged in parallel and evenly. But shortages occur for various reasons. A probability distribution is better at representing the number of bars.



Random sampling (real world): a hat with 6 types of poker chips.

- A hat has 100 poker chips by color: 5% with 1, 15% with 2, 20% with 3, 30% with 4, 20% with 5, and 10% with 6. Place the chips in a hat and shake it so that the chips are randomly distributed. Without looking, reach in and take out a chip. Write down the number on the chip. Put the chip back in the hat (with replacement), shake it, and repeat until enough samples have been collected. Note: *Computer simulations do not have hats and tokens.*

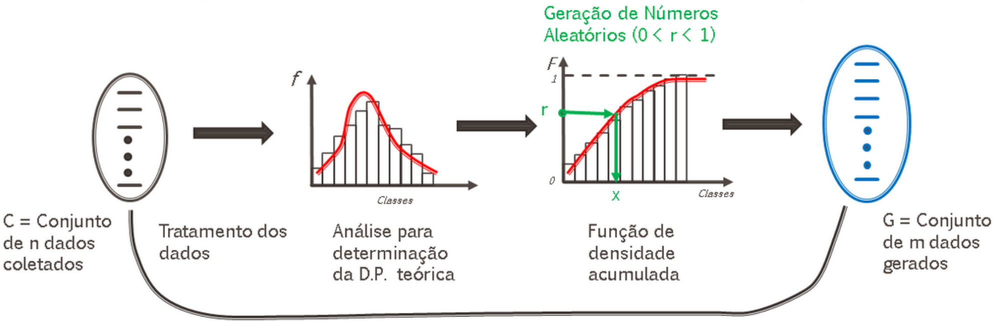




Life	Random numbers
1	0.00 to 0.04
2	0.05 to 0.19
3	0.20 to 0.39
4	0.40 to 0.69
5	0.70 to 0.89
6	0.90 to 0.99

Random sampling in computer simulation.

How are random variable values generated in a simulation?



- Data points and values differ in sets **C** and **G**. *What do they have in common?* They behave the same, i.e., they follow the same probability distribution.

Random Numbers vs. Pseudo-Random Numbers

- Pseudo-random numbers are generated by mathematical rules and pass all randomness tests, so they can be used as if they were random numbers. Mathematical functions must produce identical sequences of pseudo-random numbers to ensure test conditions are equivalent. Let:

$$\begin{aligned} x_i &\equiv \text{sequence of integers in interval } (0, m-1) \\ a &\equiv \text{constant multiplicative} \\ c &\equiv \text{constant additive} \\ m &\equiv \text{module (the remainder of a division. Notation: } \%) \\ x_0 &\equiv \text{seed} \end{aligned}$$

- Congruential generators: $x_{i+1} = (ax_i + c) \ \% \ m$
- Congruential multiplicative ($c = 0$) generators: $x_{i+1} = (ax_i) \ \% \ m$
- **Requirements**: Numbers uniformly distributed in the range [0,1] independently. Maximal cycle length and fast generation process.

Random Numbers vs. Pseudo-Random Numbers

Example: $x_0 = 4, \quad a = 5, \quad c = 0, \quad m = 7$ $\qquad x_{i+1} = (5x_i)mod(7)$

i	x_i	R_i
0	4	0.5714
1	6	0.8571
2	2	0.2857
3	3	0.4286
4	1	0.1428
5	5	0.7143
6	4	0.5714
7	6	0.8571
8	2	0.2857
9	3	0.4286
10	1	0.1428

$$x_0 = 4$$
$$x_1 = 5 * 4 \quad mod(7) = 6$$
$$x_2 = 5 * 6 \quad mod(7) = 2$$
$$x_3 = 5 * 2 \quad mod(7) = 3$$
$$x_4 = 5 * 3 \quad mod(7) = 1$$
$$x_5 = 5 * 1 \quad mod(7) = 5$$
$$x_6 = 5 * 5 \quad mod(7) = 4$$

There is periodicity

Valor maximo de $x_i = m - 1$

$$R_i = \frac{x_i}{m}, \qquad 0 \leq R_i \leq 1$$

Graphical tests: Pseudo-Random Numbers [Pidd, 1998]

Examples: (Access: https://github.com/joaoflavioufmg/sls/tree/main/06)
Run File: `ex1-pseudo-random-numbers-generator.py`

1. $x_{i+1} = (19031x_i + 9298x_{i-1}) \ \% \ 65536$
2. $x_{i+1} = 16807x_i \ \% \ (2^{31} - 1)$
3. Numpy library (Python)

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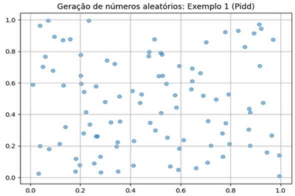
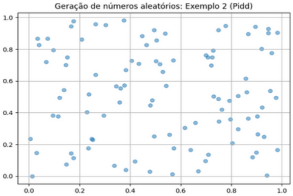
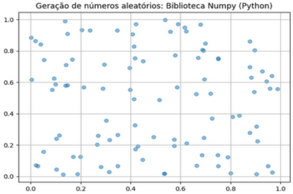
Graphical test: N = 100 (*Are the 3 good generators? What if N = 10,000?*)







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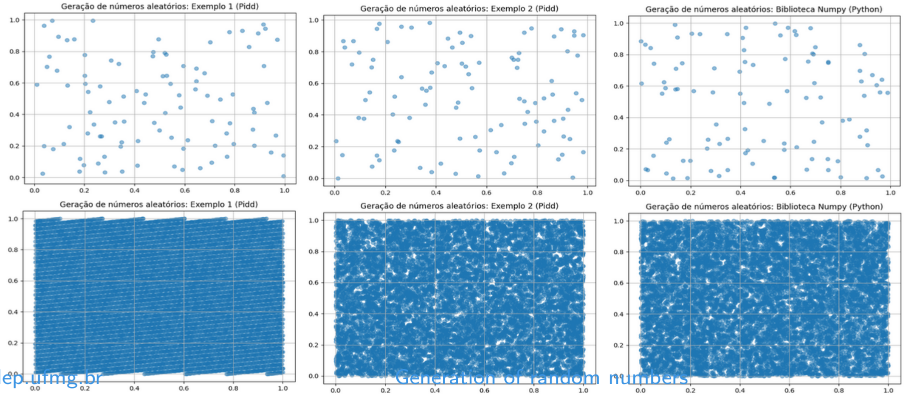

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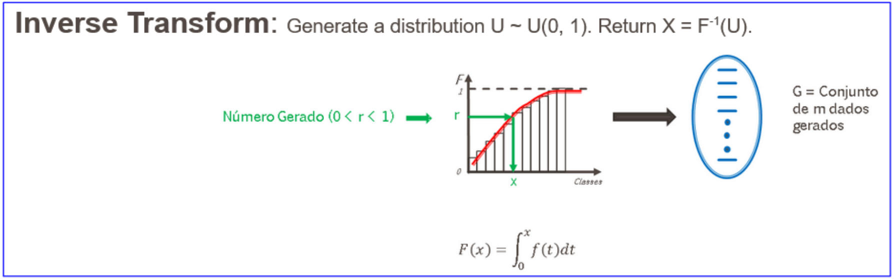
Generating random variables

- A stochastic simulation involves generating **random variables** from probability distributions. How does the simulation use random variables to run? It's based on uniformly distributed random numbers (0, 1). So, it assumes **a good random number generator** is available, and algorithm accurate, efficient, low storage, high speed, low complexity, and robust.



Inverse Transform: Exponential Distribution [Pinto, 2016]

- Exp($1/\lambda$). Mean: $m = 1/\lambda$. Std. Dev: $1/\lambda$
- $f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$
- $F(x) = \int_0^x f(t)dt = \int_0^x \lambda e^{-\lambda t} dt$
- $F(x) = \begin{cases} 1 - e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$

1. Calculate the CDF for x: $F(x) = 1 - e^{-\lambda x}, x \geq 0$
2. Replace F(x) with r, (a random number generated), and solve the equation in terms or r:
   - $r = 1 - e^{-\lambda x}$
   - $e^{-\lambda x} = 1 - r$
   - $-\lambda x = ln(1 - r)$
   - $x = -1/\lambda \; ln(1 - r)$
   - $x = -m \; ln(1 - r)$
3. Each value of r generate a random variable x: $x_i = -m \; ln(1 - r_i)$

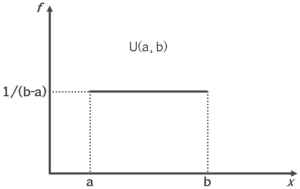
Inverse Transform: Uniform Distribution [Pinto, 2016]

- $f(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0 \end{cases}$
- $F(x) = \int_{-\infty}^{x} f(t)dt = \int_{-\infty}^{x} \frac{1}{b-a}dt$
- $F(x) = \begin{cases} 0, & x < a \\ \frac{x-a}{b-a}, & a \leq x < b \\ 1, & x \geq b \end{cases}$

1. Calculate the CDF for x: $F(x) = \frac{x-a}{b-a}, a \leq x < b$
2. Replace F(x) with r, (a random number generated), and solve the equation in terms or r:
   - $r = \frac{x-a}{b-a}$
3. Each value of r generate a random variable x: $x_i = a + (b-a)r_i$

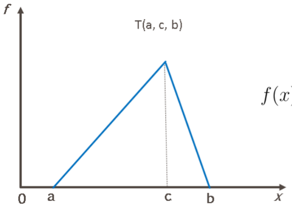
Random distributions

- **Uniform**: A distribution where values are evenly spread between two extreme values, a and b. It's a good estimate when you know only the limits of a variable.



$$f(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0 & \end{cases}$$

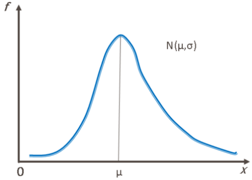
- **Triangular**: Continuous distribution for events with little data and known minimum, maximum, and likely values.



$$f(x) = \begin{cases} \frac{2(x-a)}{(b-a)(c-a)} & \text{para } a \leq x \leq c \\ \frac{2}{(b-a)} & \text{para } x = c \\ \frac{2(b-x)}{(b-a)(b-c)} & \text{para } c \leq x \leq b \\ 0 & \text{para outros casos} \end{cases}$$

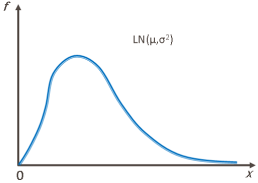
Random distributions

- **Normal**: A common distribution used to model sums of other quantities.



$$f(x) = \frac{1}{\sigma\sqrt{2\pi})} e^{\frac{1}{2}(\frac{x-\mu}{\sigma})^2} \quad \text{para} - \infty < x < \infty$$

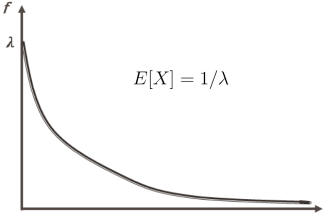
- **LogNormal**: Continuous distribution used to model intervals between failures and equipment maintenance times.



$$f(x) = \frac{1}{x\sigma\sqrt{2\pi}} e^{\left[\frac{-(ln(x)-\mu)^2}{2\sigma^2}\right]} \quad \text{para} \quad x > 0$$

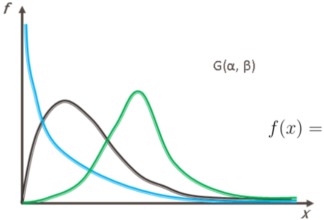
Random distributions

- **Exponential**: A common model for processes involving arrival of entities.



$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{para } x \geq 0 \\ 0 & \text{para } x < 0 \end{cases}$$

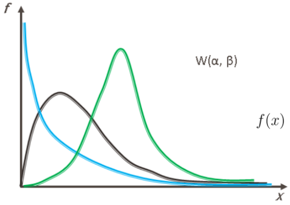
- **Gamma**: A common task completion time distribution.



$$f(x) = \begin{cases} \beta^{\alpha} x^{\alpha-1} e^{-\beta x} & \text{para } x, \alpha, \beta > 0 \\ 0 & \text{para } x < 0 \end{cases}$$

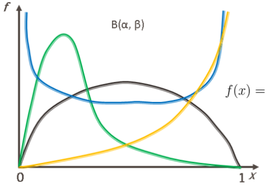
Random distributions

- **Weibull**: A common distribution used to model equipment failure intervals.



$$f(x) = \begin{cases} \alpha\beta^{-\alpha}x^{\alpha-1}e^{-(x/\beta)^{\alpha}} & \text{para } x \geq 0, \alpha, \beta > 0 \\ 0 & \text{para } x < 0 \end{cases}$$

- **Beta**: A common model of variables limited to interval (0, 1), but can represent variables in other intervals.



$$f(x) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}x^{\alpha-1}(1-x)^{\beta-1} \qquad \text{para } 0 < x < 1,\ \alpha, \beta > 0$$

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Random distributions Adherence Tests

After collecting and processing the data, the question remains: *Which probability distribution best fits the data?*

1. Construct a sample histogram.
2. Perform adherence tests to verify that the data fit a theoretical probability distribution adequately.

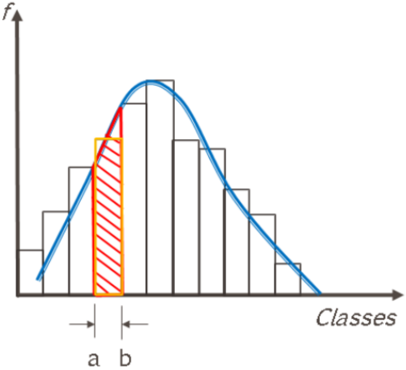
The classic and most common tests are:

- Chi-square test
- Kolmogorov-Smirnov test.

Chi-square test

Compute deviations between the frequencies of observed classes and those predicted in the theoretical distribution.
Let:

- TD_c: Theoretical data for class C:
$p_c = \int_a^b f(x)dx$, $TD_c = p_c N$
- RD_c: Real (observed) data for class C:
- E_c: Deviation between the observed and the theoretically expected:
$E_c = \frac{(RD_c - TD_c)^2}{TC_c}$, $ET = \sum_c E_c$



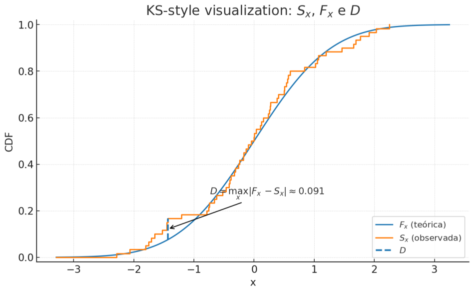
- ET has a chi-square distribution with k-1-n degrees of freedom and a ET^* with a significance level of α. Parameters from the sample: n, and k classes in the histogram. Each class must have at least 5 points, if not, classes must be grouped. If $ET < ET^*$, the hypothesis that the sample comes from a population with the adopted theoretical distribution is rejected. Otherwise, it is accepted.

Kolmogorov-Smirnov test

If the sample size is small, it is better to use the K-S test, as the Q-Q test may not be conclusive. The K-S test is based on a comparison between the theoretical and observed cumulative distributions. Let:

- D: is the maximum absolute distance;
- S_x is the observed cumulative function;
- F_x is the theoretical cumulative function.

$$D = max_x|F_x - S_x|$$



- The smaller the value of D, the closer the two cumulative functions (S_x, F_x) will be to each other. In this course, adherence tests are implemented in a programming routine in Python to be used in discrete event simulation with SimPy.

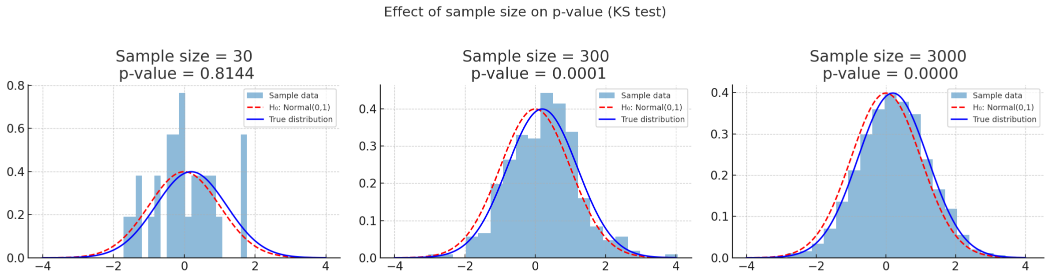
Interpreting p-value

- The p-value, (of a fit test), shows what's most likely true given the data, given the null hypothesis. It is the lowest possible level of significance (e.g. 5%) for rejecting the null hypothesis (H_0). Thus:
  - If $p \leq \alpha$, we reject H_0.
  - If $p > \alpha$, we do not reject H_0.

The lower the p-value, the greater the evidence against H_0. But be careful: p-values are sensitive to sample size. In very large samples, small differences can generate very low p-values. In small samples, large differences may not be detected.

Interpreting p-value

- **Null hypothesis** (H_0): Data follows normal distribution $\mathcal{N}(0,1)$ (red curve).

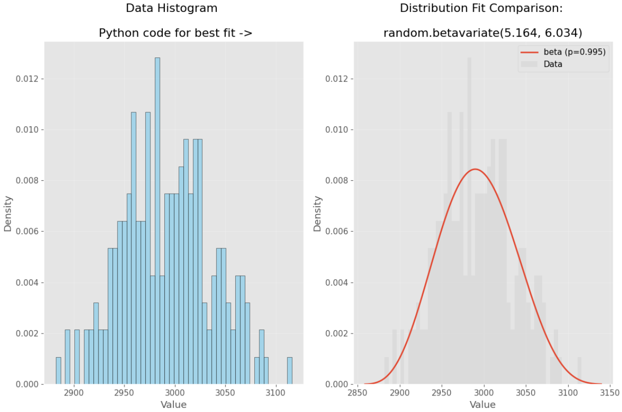


The true distribution (blue) is slightly different from the null hypothesis distribution (red dashed). As the sample size grows:

- $n = 30$: p-value high (no evidence to reject H_0);
- $n = 300$: p-value smaller (moderate evidence against H_0);
- $n = 3000$: p-value ≈ 0 (strong evidence against H_0).

Input Analyzer in Python: (https://github.com/joaoflavioufmg/sls/)

```
(venv) PS C:\...\sls\06\input> python input.py -d entrada12.txt
```



Top 3 Distributions by P-value:

1. Beta: p-value = 0.9946 (Significant: Yes)
2. Weibull: p-value = 0.9785 (Significant: Yes)
3. Lognorm: p-value = 0.9779 (Significant: Yes)

References

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Pinto, L. R. (2016).
Simulação por eventos discretos.
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